Practical DFT
Introduction to GPAW

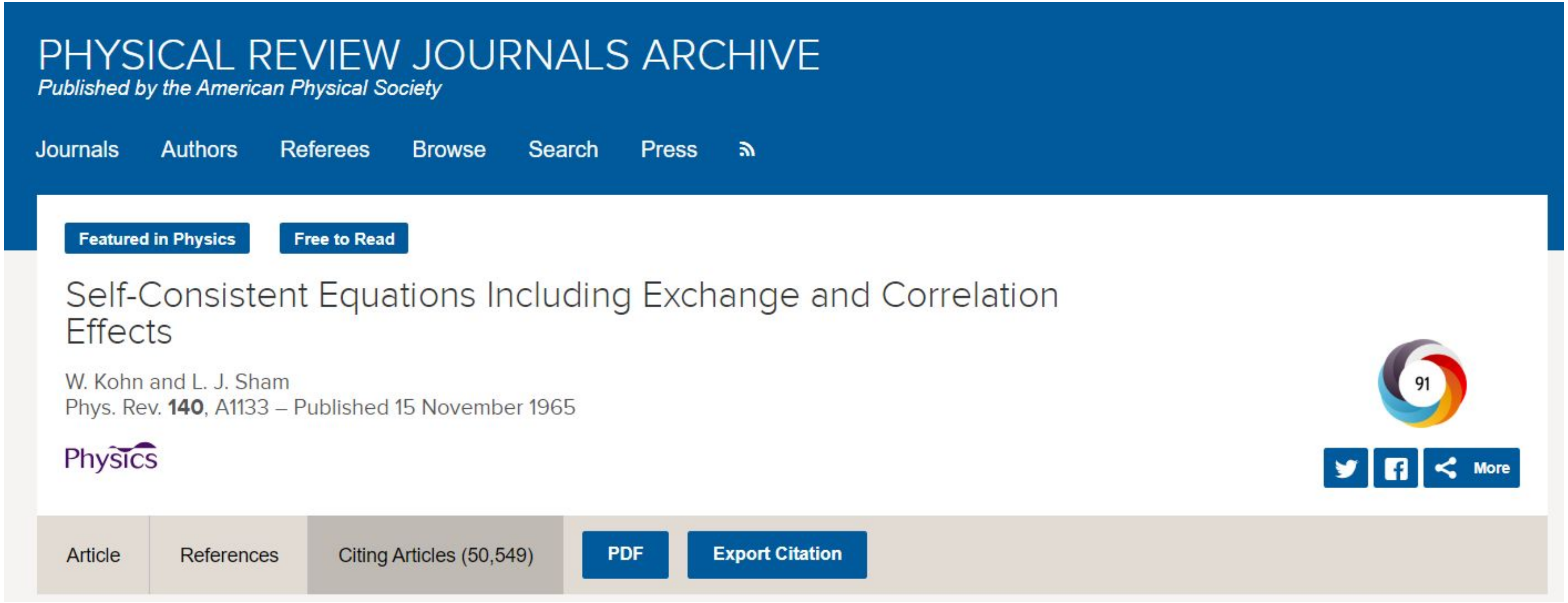
10/09/2024

Outline

1. Motivation
2. Visualization with Vesta
3. Introduction to Linux (on Windows or Mac)
4. Introduction to conda environments
5. Getting started with Jupyter notebooks
6. First GPAW calculations

Motivation

- Covered theoretical/mathematical foundations of DFT
- Note how many papers cited the Kohn Sham paper (1965) to date, enabled by advances in computational power → *e.g.* check out Google Scholar of VM



Many DFT codes

VASP: Vienna Ab-initio Simulation Package

QUANTUM ESPRESSO

CASTEP

GPAW!

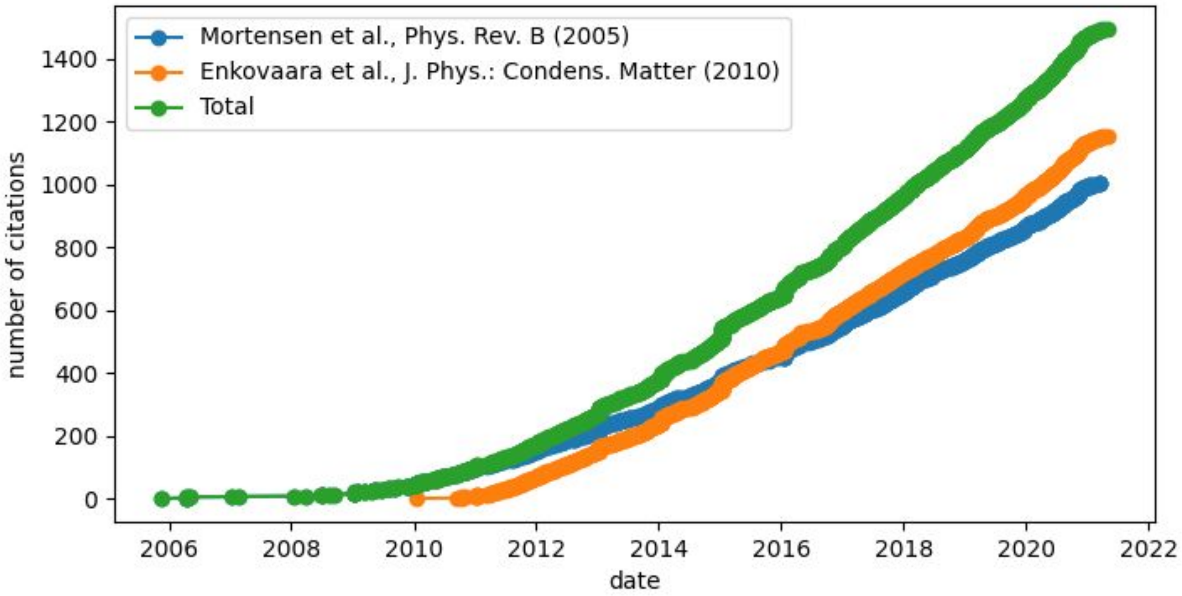
Abinit

siesta

CP2K

DFT with GPAW

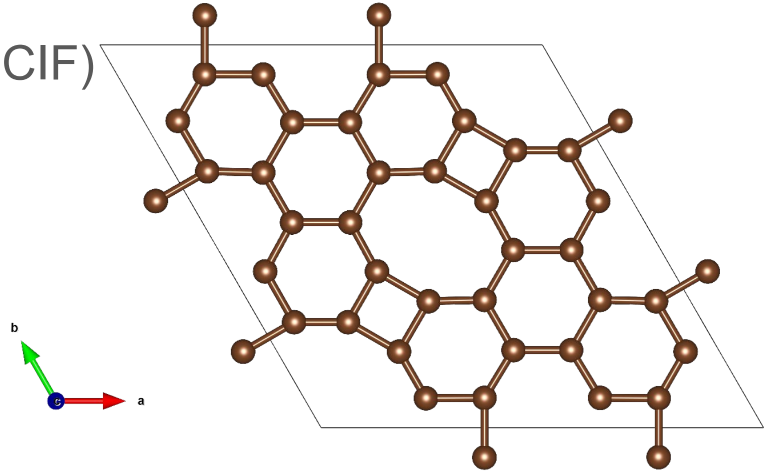
- What is GPAW? In short: Python package for electronic structure calculations



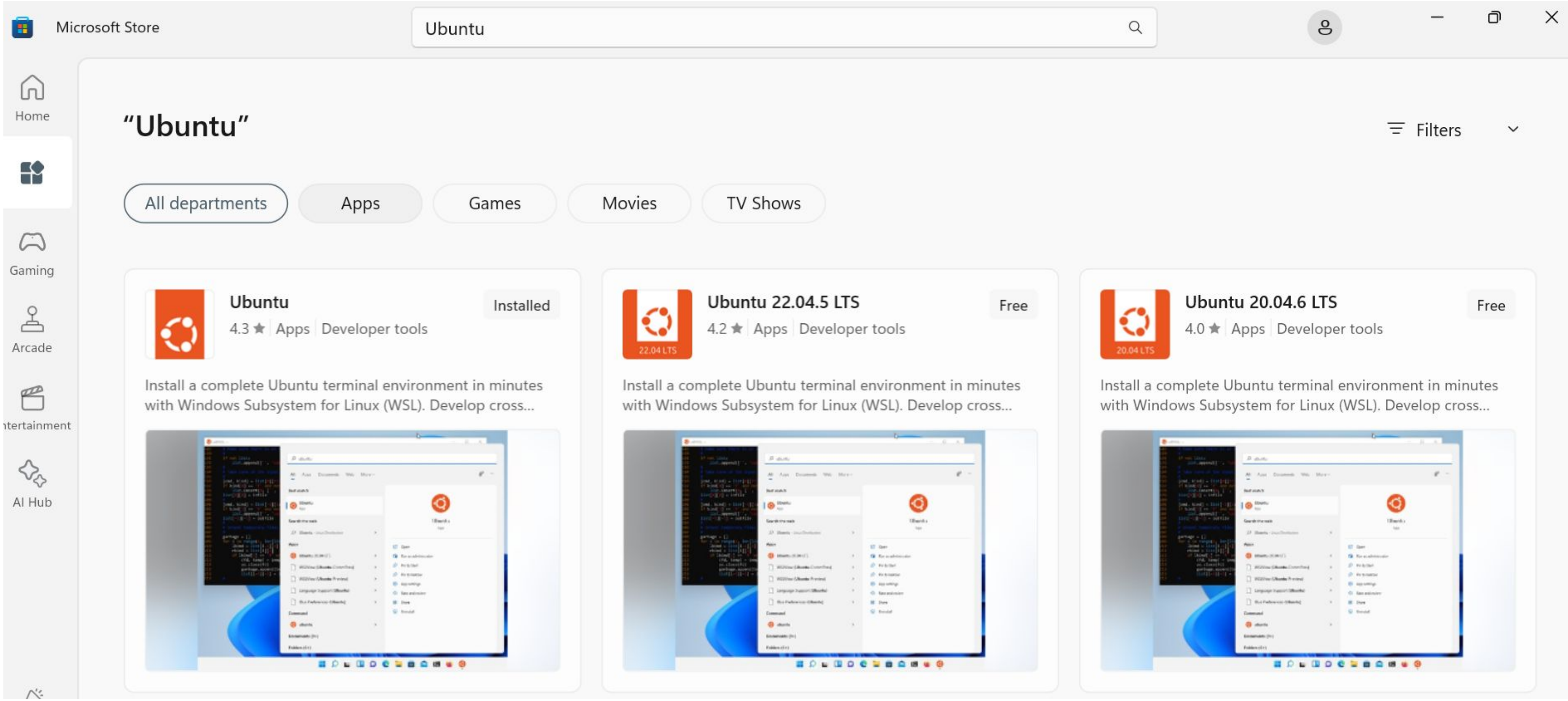
- https://jensj.gitlab.io/gpaw-2021-talk/index.html
- https://gpaw.readthedocs.io/faq.html

Hands-on Exercise: Vesta & Your first Structure File

- Got to https://jp-minerals.org/vesta/en/download.html
- Get the installer for your operating system & follow installation instructions
- Get the structure file from the Zoom chat and drag it into the Vesta window
- Explore the visualization options (save as CIF)



Introduction to Linux: Install Ubuntu WSL on Windows



Introduction to Linux: Use Terminal on macOS

What is Terminal on Mac?

Terminal is an app for advanced users and developers that lets you communicate with the Mac operating system using a command line interface (CLI). You enter commands and scripts (called *shell scripts*) to perform tasks on your Mac.

https://support.apple.com/guide/terminal/welcome/mac

Some basic terminal commands

- Navigate your computer's file system along with base-level tasks such as create, copy, rename, and delete:
  - Move around your directory structure: `cd`
  - Create directories: `mkdir`
  - Create files (and modify their metadata): `touch`
  - Copy files or directories: `cp`
  - Move files or directories: `mv`
  - Delete files or directories: `rm`

- https://developer.mozilla.org/en-US/docs/Learn/Tools_and_testing/Understanding_client-side_tools/Command_line
- https://medium.com/@prasku/basic-terminal-commands-ce5790c20107

Conda & Conda Environments

- Conda provides package, dependency, and environment management for any language.
- https://docs.anaconda.com/miniconda/

Miniconda

Miniconda is a free minimal installer for conda. It is a small bootstrap version of Anaconda that includes only conda, Python, the packages they both depend on, and a small number of other useful packages (like pip, zlib, and a few others).

If you need more packages, use the `conda install` command to install from thousands of packages available by default in Anaconda's public repo, or from other channels, like conda-forge or bioconda.

Conda & Conda Environments

Quick command line install

These quick command line instructions will get you set up quickly with the latest Miniconda installer. For graphical installer (.exe and .pkg) and hash checking instructions, see Installing Miniconda.

Note

For best results, copy all of the commands in each code block and run them all at once.

Windows Command Prompt | Windows PowerShell | macOS | **Linux**

These four commands download the latest 64-bit version of the Linux installer, rename it to a shorter file name, silently install, and then delete the installer:

```
mkdir -p ~/miniconda3
wget https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86_64.sh -O ~/miniconda
bash ~/miniconda3/miniconda.sh -b -u -p ~/miniconda3
rm ~/miniconda3/miniconda.sh
```

- https://docs.anaconda.com/miniconda/
- https://learning.anaconda.cloud/conda-basics

Conda & Conda Environments

Managing environments

With conda, you can create, export, list, remove, and update environments that have different versions of Python and/or packages installed in them. Switching or moving between environments is called activating the environment. You can also share an environment file.

```
conda create -n myenv python
```

Removing an environment

To remove an environment, in your terminal window, run:

```
conda remove --name myenv --all
```

https://docs.conda.io/projects/conda/en/latest/user-guide/tasks/manage-environments.html

GPAW In your Conda Environment

```
conda create -n gpaw_env python=3.12
conda activate gpaw_env
conda list
conda install conda-forge::gfortran
conda install conda-forge::gpaw

gpaw test

conda install ipykernel

conda deactivate  [This deactivates the environment]
```

Important: GPAW is not compatible with the newest Python version (3.13), so please make sure to specify the python version when creating the environment

Python in Jupyter Notebooks

- Now that we’re outside the gpaw environment, (it should say base environment), do:

conda install jupyter

Run in terminal:

jupyter-notebook &

Copy the link into your browser

Free software, open standards, and web services for interactive computing across all programming languages

First ASE and GPAW calculation

→ Let’s go over to the jupyter-notebook and try out things in real time

More Notebooks To Try Here:
https://gpaw.readthedocs.io/summerschools/summerschool24/intro/intro.html